STAT405 Project Report

Bayesian Study of Purchase Probability in Online Shopping Sessions

Github Link: https://github.com/lkwlsleyw/STAT405_finalProject

1. Introduction

Online shopping conversion is important because it shows whether a browsing session ends with a purchase. This matters for customer behavior analysis, website design, and business decisions. In this project, I study purchase probability using Bayesian inference. Instead of treating the problem only as a classification task, I focus on uncertainty in the estimated effects and in the predicted purchase probabilities.

The main question is: **which session-level variables are related to purchase probability, and does a hierarchical Bayesian model with month-specific intercepts fit the data better than a simpler baseline model?**

This project compares two Bayesian logistic regression models. It also compares two posterior computation methods for the baseline model and uses model checks to evaluate fit.

2. Literature review

The Online Shoppers Purchasing Intention Dataset was introduced by Sakar et al. (2018), who used session-level behavioral variables for real-time prediction of purchase intention. This shows that the dataset is usually used for purchase prediction.

More generally, session-level purchase intent in e-commerce has often been studied as a prediction problem. For example, Bogina et al. (2019) studied whether an anonymous browsing session would end with a purchase using session information. Their work suggests that session behavior contains useful signals for conversion.

This project differs from the most closely related work in two main ways. First, it focuses on Bayesian uncertainty quantification rather than only prediction accuracy. Second, it compares a baseline Bayesian logistic regression with a hierarchical extension that allows month-level variation. This idea is also related to the broader marketing literature, where Allenby et al. (2005) discussed the value of hierarchical Bayes when there is group-level heterogeneity.

3. Data and preprocessing

This project uses the Online Shoppers Purchasing Intention Dataset, which contains 12,330 sessions and a binary outcome variable, Revenue. I kept a smaller set of interpretable variables: Revenue, Month, PageValues, BounceRates, ExitRates, ProductRelated_Duration, VisitorType, and Weekend. There were no missing values in the final analysis data.

Revenue and Weekend were encoded as binary 0/1 variables. VisitorType was simplified into a binary variable called ReturningVisitor, where 1 means Returning_Visitor and 0 means otherwise. Month was converted into an integer grouping variable called MonthIndex. The dataset contains 10 observed months, so the hierarchical structure is based on the months that appear in the data. The continuous predictors PageValues, BounceRates, ExitRates, and ProductRelated_Duration were standardized to mean 0 and standard deviation 1.

4. Descriptive summary

The overall purchase rate in the data is 0.1547, so about 15.5% of sessions ended in a purchase. Purchase rates vary across months, from about 0.016 in February to about 0.254 in November (see Figure A1 in the Appendix), which gives some support for a month-varying intercept model. The raw purchase rate is 0.246 for non-returning sessions and 0.139 for returning sessions. The continuous predictors also show reasonable patterns: sessions with purchases have higher PageValues and ProductRelated_Duration, but lower BounceRates and ExitRates on average.

5. Models

I compare two Bayesian logistic regression models for the purchase outcome.
Let Y_i be the purchase outcome for session i, where $Y_i = 1$ means a purchase happened and means no purchase happened. In both models, $Y_i \sim Bernoulli(\theta_i)$, where θ_i is the purchase probability for session i.

Model 1: baseline Bayesian logistic regression

$$logit(\theta_i) = \alpha + \beta_1 PagesValue_i + \beta_2 BounceRates_i + \beta_3 ExitRates_i + \beta_4 ProductRelatedDuration_i + \beta_5 ReturningVisitor_i + \beta_6 Weekend_i$$

I used the priors $\alpha \sim Normal(0, 2)$, and $\beta_i \sim Normal(0, 1.5),\ i = 1, ..., 6$

This model assumes one common baseline intercept for all sessions.

Model 2: hierarchical Bayesian logistic regression

The second model allows the intercept to vary by month:

$$logit(\theta_i) = \alpha_{m[i]} + \beta_1 PagesValue_i + \beta_2 BounceRates_i + \beta_3 ExitRates_i + \beta_4 ProductRelatedDuration_i + \beta_5 ReturningVisitor_i + \beta_6 Weekend_i,$$ where $m[i]$ is the month group for session i.

The month-specific intercepts follow: $\alpha_m \sim Normal(\mu_\alpha, \sigma_\alpha)$,
with hyperpriors $\mu_\alpha \sim Normal(0, 2),\ \sigma_\alpha \sim Exponential(1)$.

This model allows the baseline purchase tendency to differ across months while still sharing information across groups.

6. Posterior computation and checks

For the baseline model, I used both Laplace approximation and HMC in Stan. For the hierarchical model, I used HMC in Stan.
The prior predictive check showed a wide range of possible mean purchase rates before seeing the data. The simulated values ranged from about 0.006 to 0.993, with median around 0.509. This means the priors were weak and allowed many possible outcomes.

For the HMC fits, I ran 4 chains. Both models had zero divergent transitions and zero maximum treedepth warnings. The $\widehat{R}$ values were all very close to 1, and the effective sample sizes were large for the main parameters. This suggests that the chains mixed well and the posterior summaries are reliable.

I also used a small correctness check. I simulated data from the baseline model with known parameter values, then fitted the same model back to the simulated data. The posterior means were close to the true values, and the true values were inside the posterior intervals for all parameters.

7. Results

7.1 Baseline model: Laplace vs HMC
The baseline model gave clear evidence for several important effects.

PageValues_std had a strong positive association with purchase probability. The posterior mean was about 1.48 under both Laplace and HMC. ExitRates_std had a strong negative association, with posterior mean about -0.82 under both methods. ProductRelated_Duration_std had a positive effect of about 0.27. ReturningVisitor had a negative coefficient of about -0.32, and Weekend had a small positive coefficient of about 0.15. BounceRates_std was negative on average, but the uncertainty was larger and the interval still included values near zero, so this effect looks weaker than the others.

A main result is that Laplace approximation and HMC gave very similar results for Model 1. For example, the posterior mean for PageValues_std was 1.481 under Laplace and 1.483 under HMC. For ExitRates_std, the values were -0.820 and -0.825. The posterior standard deviations were also very similar across the two methods. This suggests that, for the baseline model, the posterior is regular enough that a normal approximation works reasonably well. In this case, Laplace approximation gives almost the same practical conclusions as HMC for the main coefficients, so it seems to be a useful and much simpler approximation for this model.

Table1: *Model 1 Laplace vs HMC summary*

	parameter	laplace_mean	laplace_sd	laplace_lower95	laplace_upper95	hmc_mean	hmc_sd	hmc_lower95	hmc_upper95
1	alpha	-1.9621940	0.08971097	-2.13802752	-1.7863605	-1.9711298	0.08699279	-2.11350017	-1.8284723
2	PageValues_std	1.4805381	0.04315194	1.39596031	1.5651159	1.4833525	0.04282830	1.41257457	1.5558609
3	BounceRates_std	-0.1202149	0.15119363	-0.41655441	0.1761246	-0.1262435	0.15309959	-0.38275385	0.1238745
4	ExitRates_std	-0.8200096	0.11101646	-1.03760186	-0.6024173	-0.8254869	0.11293262	-1.01348691	-0.6439332
5	ProductRelated_Duration_std	0.2746074	0.02595059	0.22374426	0.3254706	0.2745428	0.02530916	0.23363126	0.3157776
6	ReturningVisitor	-0.3242732	0.08283778	-0.48663523	-0.1619111	-0.3209653	0.08360177	-0.45918827	-0.1786049
7	Weekend	0.1484886	0.06958326	0.01210541	0.2848718	0.1475016	0.06953883	0.03408343	0.2627701

7.2 Hierarchical model
The hierarchical model gave similar fixed-effect results, but it also captured clear month-to-month differences in baseline purchase tendency.

The main fixed effects stayed stable. PageValues_std remained strongly positive, with posterior mean about 1.53. ExitRates_std remained strongly negative, around -0.79. ProductRelated_Duration_std stayed positive, around 0.20. ReturningVisitor remained negative, around -0.30. The effect of Weekend became weaker in the hierarchical model, with its interval including values close to zero. BounceRates_std again looked negative on average, but it still had more uncertainty than the strongest predictors.

The hierarchical part also showed meaningful month-level variation. The largest month intercept was for November (about -1.32), while the smallest was for February (about -2.89). Since a higher intercept means a higher baseline purchase probability on the log-odds scale, this agrees with the descriptive result that November had much higher conversion than February.

The posterior mean of σ_α was about 0.53, suggests a moderate amount of month-level heterogeneity.

Table 2: *Model 2 main posterior summary*

	parameter	mean	sd	lower95	upper95	rhat	ess_bulk	ess_tail
1	mu_alpha	-2.0871772	0.19752396	-2.40174798	-1.77240264	1.003955	1156.627	1920.567
2	sigma_alpha	0.5313815	0.17221127	0.31739788	0.86508260	1.001926	1206.549	1855.735
3	PageValues_std	1.5283305	0.04467298	1.45558244	1.60192966	1.000656	4199.160	2731.880
4	BounceRates_std	-0.1860595	0.15681377	-0.44160677	0.07072054	1.002422	3078.932	3192.052
5	ExitRates_std	-0.7941145	0.11164384	-0.97883779	-0.61271106	1.003064	2896.907	3060.274
6	ProductRelated_Duration_std	0.2015158	0.02503267	0.15942513	0.24279687	1.001589	3855.322	2977.379
7	ReturningVisitor	-0.3014234	0.08466303	-0.43853352	-0.15841423	1.000087	3432.065	2598.587
8	Weekend	0.1042720	0.06962545	-0.01290881	0.21707301	1.000567	3929.698	2769.032

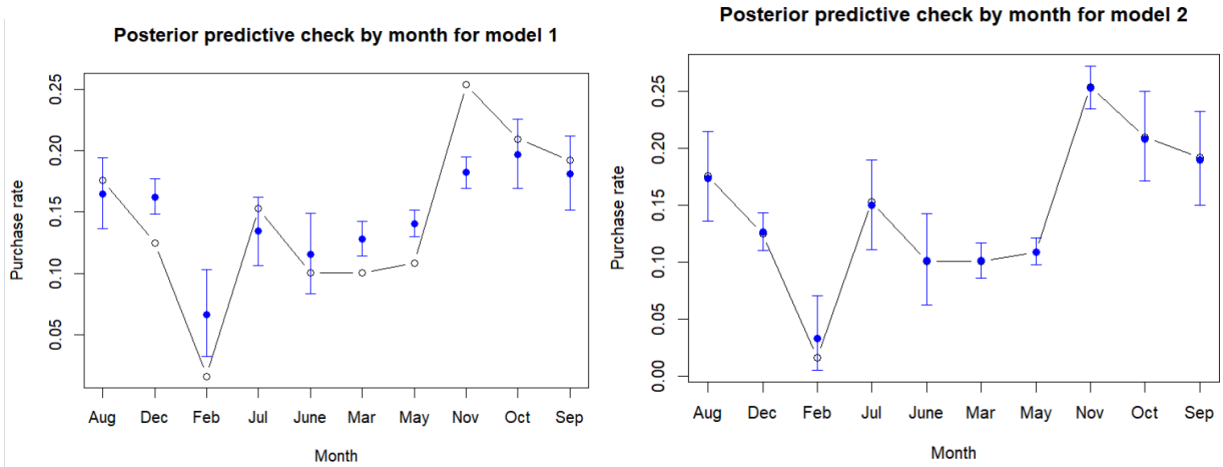
7.3 Posterior predictive checks

The overall posterior predictive check was acceptable for both models. The observed overall purchase rate was 0.1547, and both models produced posterior predictive means close to this value.

The month-level posterior predictive check showed a clearer difference between the two models. For Model 1, the predicted purchase rates were too shrunk toward the middle. For example, it overestimated February and underestimated November. In contrast, Model 2 matched the observed month pattern much better. Its predicted means were much closer to the observed values for both low-conversion and high-conversion months.

So, both models fit the overall purchase rate well, but the hierarchical model gave a noticeably better fit to the month-level structure.

Figure2: *posterior predictive check by month for Model1&2*



8. Discussion

Overall, the results suggest that PageValues, ExitRates, and ProductRelated_Duration are the most important session-level predictors in this analysis. The month-varying intercept model also seems useful, because it captures the monthly differences much better than the baseline model.

There are still some limitations. First, this is an observational dataset, so the coefficients should not be interpreted as causal effects. Second, the month grouping is helpful, but it is still a fairly simple hierarchical structure. A richer grouping variable, such as user-level or campaign-level information, would probably be more informative if it were available. Third, the prior predictive check showed that the priors were quite broad, so a slightly tighter prior could also be reasonable in future work.

Even with these limitations, this project shows that the dataset can be used for a meaningful Bayesian analysis, not only for black-box prediction. The baseline model was already informative, and the hierarchical model improved the fit by capturing monthly heterogeneity.

References

Allenby, Greg M. and Rossi, Peter E. and McCulloch, Robert E., Hierarchical Bayes Models: A Practitioners Guide (2005). https://ssrn.com/abstract=655541

Bogina, V., Mokryn, O., & Kuflik, T. (2019). Will this session end with a purchase? Inferring current purchase intent of anonymous visitors. Electronic Commerce Research and Applications, 34, 100836. https://doi.org/10.1016/j.elerap.2019.100836

Sakar, C.O., Polat, S.O., Katircioglu, M., & Kastro, Y. (2018). Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks. Neural Computing and Applications, 31, 6893 - 6908.

Appendix

Data source: The dataset used in this project is the Online Shoppers Purchasing Intention Dataset from the UCI Machine Learning Repository. Online Shoppers Purchasing Intention Dataset - UCI Machine Learning Repository

Code files are available in the github repository.

AI is used for debugging code, finding relevant sources, suggesting ideas for models, checking grammar and writing reference citation.

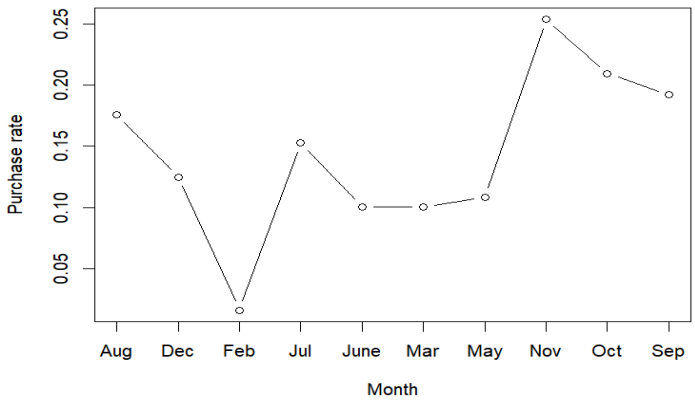



Figure A1: *Purchase rate by month*